

SagarDrishti

A browser-native 3D digital twin of India's ocean, built for the Ministry of Earth Sciences and INCOIS.

The whole project from zero. It assumes you have never heard of INCOIS, an Argo float or a NetCDF file, and ends with you knowing what we built, why it did not exist before, and how each part works.

01

The whole water column drawn as a volume in a browser, with the real instruments positioned inside it.

02

A live score, by the international Class-4 method, of how far the simulation is from those instruments.

03

India's disaster warnings drawn over the water they concern, in the CAP standard the government already speaks.

04

A question box that cannot invent a number, because every figure arrives as a tool result with its citation.

PROBLEM OWNER

Ministry of Earth Sciences,
INCOIS Hyderabad

CATEGORY

Software · Disaster
Management

TEAM

PixelPaws

DOCUMENT DATE

20 September 2026

The problem

No background assumed. Every term used later is introduced here, and the glossary on the last page repeats them.

1.1 Who INCOIS is

The **Indian National Centre for Ocean Information Services**, Hyderabad, under the **Ministry of Earth Sciences**. It is the body that tells India what its ocean is doing. It runs India's Tsunami Early Warning Centre and is a UNESCO-recognised Tsunami Service Provider for the wider Indian Ocean; it issues the daily Ocean State Forecasts behind high wave and swell surge advisories; it supports Coast Guard search and rescue, where knowing which way the water moved *at depth* decides where to look; and its Potential Fishing Zone advisories reach roughly nine lakh fishing households.

All four jobs turn on one question asked under time pressure: *what is the ocean doing right now, and do I believe it?*

1.2 The two kinds of ocean data

There are exactly two ways to know, and the gap between them is the whole problem.

A **numerical ocean model** is a physics simulation. It cuts the ocean into boxes a few kilometres wide and tens of levels deep and solves equations of motion and heat for each one. The output is complete and gap-free: a temperature, a salinity and a water velocity for every longitude, latitude, depth and time. It is also a guess. A good one, but a guess.

An **in situ observation** is a real instrument in real water reporting a real number. True, and almost empty: a handful of points in an ocean. The problem statement names four kinds.

Instrument	What it is
Argo float	A metre-long cylinder that drifts at about 1000 m, then every ten days sinks to 2000 m and rises, recording temperature and salinity the whole way up, transmitting by satellite at the surface. About four thousand are active worldwide, each with a unique WMO identifier. One rise is a profile or cast : a column of measurements at one place and time.
BGC float	The same machine with more sensors: dissolved oxygen, chlorophyll, nitrate, pH and particle backscatter.
Glider	A winged robot that changes its buoyancy to fly up and down in a sawtooth, steered from shore. It surveys a chosen line instead of drifting.
CTD cast	The oldest method. A ship stops and lowers an instrument on a wire. Conductivity, Temperature, Depth, because salinity is computed from conductivity rather than measured.

THE CENTRAL TENSION

The model is complete but possibly wrong. The observations are correct but almost empty. A forecaster issuing a cyclone advisory must hold both in mind and judge whether today's run can be trusted in the place that matters. **Supporting that judgement is the job.**

The formats. Model output and most observations arrive as **NetCDF**, a self-describing binary format for multidimensional arrays, written to the **CF Conventions** (Climate and Forecast). Older CTD work arrives as plain delimited text. A platform that wants to read what INCOIS holds must handle all of it.

1.3 What the problem statement says

SIH26067 was written by INCOIS: *"Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations."* It is unusually direct about what is wrong. In their words:

"no integrated, web-based 3D visualization platform currently exists" that renders model fields and instrument observations "in a single interactive environment"; existing tools are **"desktop-bound, support only 2D plan views, or lack the ability to co-visualize model outputs alongside instrument profiles"**; forecasters are "forced to toggle between disparate software packages"; and this **"impedes timely hazard assessment, search-and-rescue support, fishery advisories, climate monitoring."**

The complaint is not aesthetic. The claim is that the missing tool costs time in the four operations that save lives and livelihoods.

1.4 The seven requirements, which are the objectives

#	Requirement	In plain terms
F1	3D volumetric rendering	Draw the whole water column in three dimensions, not a flat map: depth slices, isosurfaces, animation through time.
F2	Instrument data overlay	Put Argo, BGC, glider and CTD on the same globe as clickable marks, each opening its depth-versus-variable profile.
F3	Multi-format ingestion	Read NetCDF and delimited text automatically; add a source without rewriting the software.
F4	Colorbar and controls	Choose variable, depth, time, colour scale and limits, layer opacity, vertical exaggeration.
F5	Web-based, scalable	Runs in a browser with nothing to install; deployable on INCOIS servers.
F6	Extensible design	A documented way to add future sensors (moorings, HF radar, ADCP) and derived products.
F7	Open standards	Serve OGC WMS and WCS, comply with CF, so other software can consume it.

The portal files it under the theme **Disaster Management**, category **Software**. That is why the project carries a warning layer as well as a viewer.

What exists, and why it is not enough

A project claiming a gap has to prove the gap. We reviewed the landscape before writing code, and one finding changed how we talk about the project.

THE CORRECTION WE MAKE FIRST

INCOIS already runs **Digital Ocean** (2020), whose launch material claims "*3D and 4D (3D in space with time animation) data visualization*" on "a georeferenced 3D Ocean." An INCOIS judge knows that sentence, so **we never claim no 3D exists at INCOIS**. Our position is narrower and defensible: Digital Ocean is a data management portal whose 3D is a georeferenced globe with draped layers and time animation. We build with it, consuming INCOIS's own services, and add what it does not do.

2.1 The landscape

System	Where	Why the problem statement still exists
Digital Ocean	INCOIS	No documented volumetric rendering, no model skill scoring, no public embedding story. A portal with globe viewing, not an analysis-grade renderer.
Live Access Server	INCOIS	Static 2D images, one field at a time, no instrument overlay.
ERDDAP, THREDDS	INCOIS	Data access layers with rudimentary graphs. These are our backend, not competitors.
SARAT, ITEWC, OSF, PFZ	INCOIS	A world class dissemination pipeline with text and 2D map front ends. The science visualisation layer is the gap.
EDITO	EU	Infrastructure first and notebook centric, Europe scoped, no coherent volumetric globe.
Copernicus MyOcean Pro	EU	Best web ocean viewer today, but its depth axis is a <i>selector</i> : one level at a time, never the shape of the column. Their own roadmap puts a 3D globe at January 2027.
Argovis	US	The strongest co-location engine, and 2D. Our methodological ancestor; we mirror its API semantics deliberately.
nullschool, Windy	Global	Surface only. Roughly ninety per cent of ocean state, including cyclone intensification and search and rescue physics, is below the pixel they draw.
Ocean Data View	Desktop	The standard for in situ sections, weak on gridded model data. A browser version, webODV, exists and is 2D-plot based.
pyParaOcean	Desktop	Our most serious technical prior art: real volume rendering on Bay of Bengal data. It is also exactly what "desktop-bound" means, a ParaView and HPC install a forecast desk officer cannot use.
FloatChat, cesium-mcp, FathomGPT	AI	Chat over Argo tables; generic camera control; natural language over ocean imagery. All real, all cited. None touches volumetric scene state or skill metrics.

2.2 The four gaps we can defend

Capability	Closest	Why they miss
1. Volumetric water column in a browser, co-registered with in situ profiles	pyParaOcean and IDV are volumetric but desktop. MyOcean and Argovis are web but 2D.	No system occupies both cells.
2. A live model skill scorecard inside the viewer	GODAE and Copernicus quality reports	Computed offline, published as PDFs. No operational viewer shows skill beside the field.
3. Language control of a <i>scientific</i> scene graph	FloatChat, cesium-mcp	Neither knows the other's domain. A tool schema spanning volumetrics and validation does not exist.
4. Indian Ocean and INCOIS native	EDITO, Copernicus	Structurally scoped to Europe.

2.3 The four rebuttals, rehearsed

"Digital Ocean already says 3D and 4D." Correct, and we build with it. It is a data management portal with a draped-layer globe. We add true volumetrics, a quantitative skill panel and language control, while consuming INCOIS's own ERDDAP and THREDDS, so zero data duplication.

"MyOcean Pro already views four dimensions." Its depth axis is a selector, one level at a time. Copernicus's own roadmap puts their 3D globe at January 2027, and even that is without volumetrics or in situ integration.

"pyParaOcean already does this." Yes, and we cite it. It is also the definition of the desktop-bound tool the problem statement complains about. We deliver a defensible subset at a URL with zero install, plus the observation comparison and language layers it has no scope for.

"Is this not nullschool for the ocean?" Those are surface particle animations. No in situ data, no export, no skill metric, no column.

What SagarDrishti is

Sagar Drishti means "ocean vision". The name echoes the SAGAR doctrine and the Deep Ocean Mission, because the audience is a ministry.

A web page you open with no installation, showing the Indian Ocean as a three-dimensional body of water rather than a flat map. The simulated field is drawn through the whole depth of the column. The real instruments floating in that water are drawn inside it as marks you can click. A panel tells you how far the simulation is from those instruments, in degrees. The live government disaster warnings are drawn over the water they concern. And a question box answers in sentences built only from numbers the software actually computed.

13

sources enabled, across 9 kinds

149

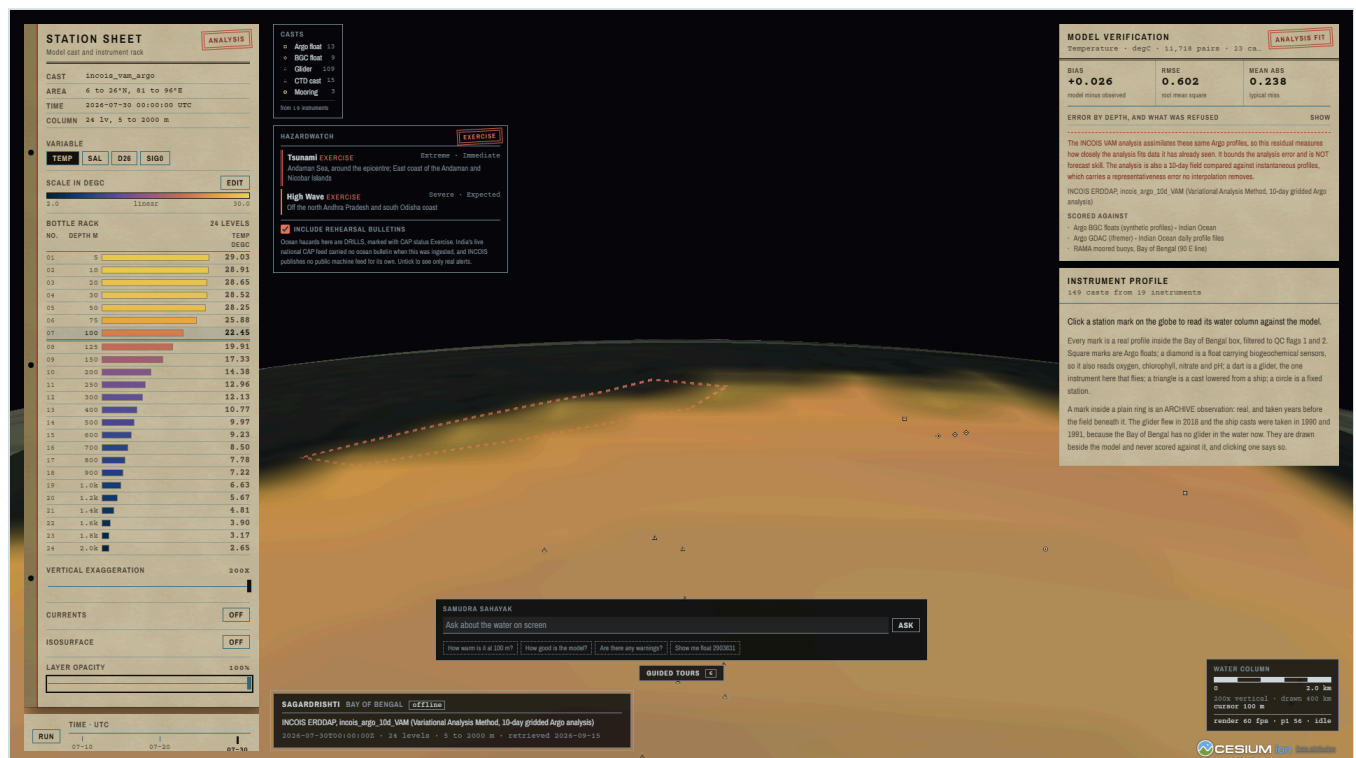
casts from 19 instruments

30,012

quality-controlled levels

567

automated tests passing



The opening view. Bay of Bengal water column in colour, instrument marks across it, a warning area outlined, and the control panel holding the variable selector, colour scale, depth ladder and vertical exaggeration. The box at top left names the dataset and timestamp on screen, so nothing is unattributed.

The core USP, in four claims

1. The water column is drawn as a volume, in a browser, with the real instruments inside it

Not a surface map with a depth dropdown. The field is rendered through depth using the three techniques the problem statement names: stacked depth slices, extracted isosurfaces, and animation through time. Argo, BGC, glider and CTD casts sit in the same 3D space, told apart by mark shape, and clicking one opens its measured profile beside the model's curve at the same place and time.



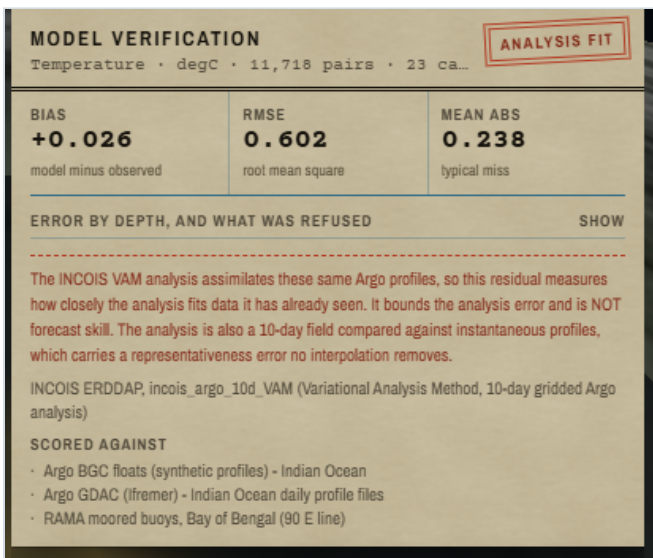
The 26 degree isosurface, enclosing all water warmer than 26 degrees Celsius, which is the layer a cyclone draws energy from. Marching cubes on the native grid, no resampling.



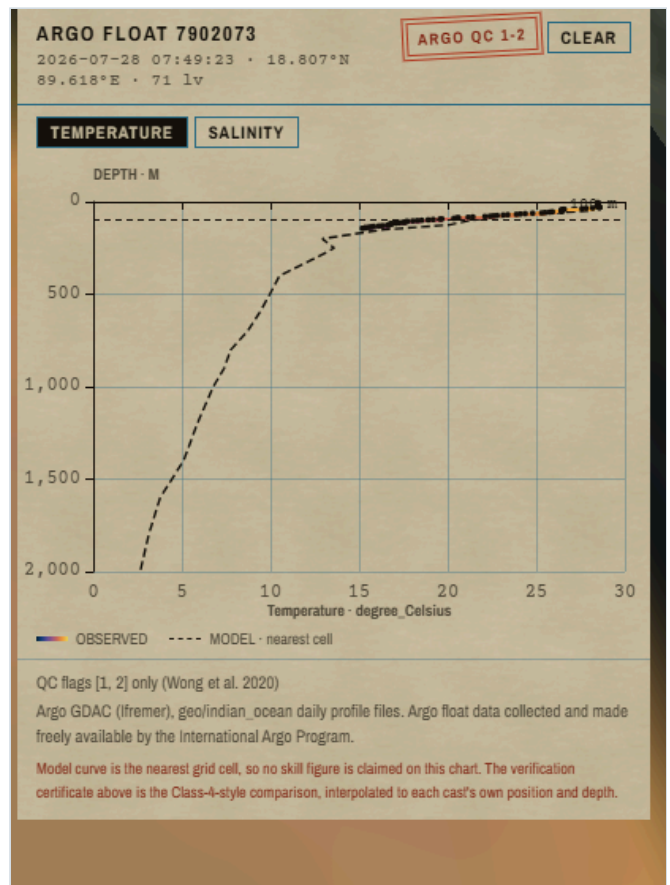
All four instrument classes the problem statement names, by mark shape rather than colour alone: Argo 13, BGC 9, glider 109, CTD 15, mooring 3.

2. The viewer tells you whether to trust what it is showing you

The differentiator a domain expert notices first. Every operational centre produces skill statistics; they are computed offline and published as quarterly PDFs. Nobody puts the number next to the picture. We do, by **GODAE OceanView Class 4**, the accepted international protocol, which works in *observation space*: the model is interpolated to each instrument's exact latitude, longitude, depth and time rather than both being averaged onto a convenient grid, and bias and RMSE are reported per depth band using Argo quality flags 1 and 2 only.



The verification card. Bias +0.026 degrees, RMSE 0.602 degrees, 11,718 matched pairs. The caveat is printed in red on the card, not hidden in documentation.



Argo float 7902073, 71 levels. Measured profile against the model's curve at the same coordinates, WMO identifier carried into every citation.

3. The disaster warnings are on the same globe as the physics

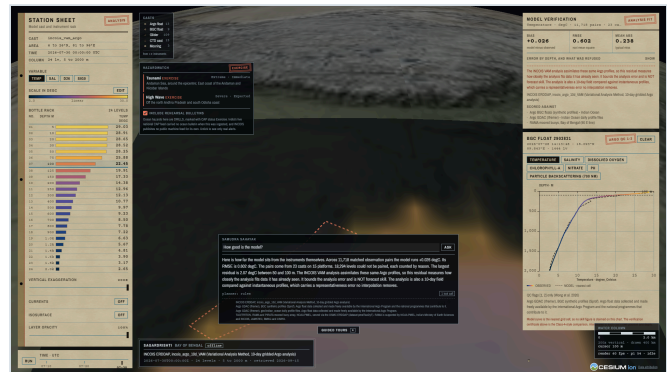
The theme is Disaster Management and the first stated mandate is "timely hazard assessment." HazardWatch reads **CAP** bulletins, the Common Alerting Protocol, which is the international XML standard for public warnings and the format India's national SACHET system speaks. Each becomes a coloured area on the water, severity driving colour, so the warning and the field explaining it are in one picture.

4. You can ask it questions, and it cannot make anything up

Samudra Sahayak, "ocean helper", is the question box. The design constraint is the interesting part: **it is structurally unable to compute or invent a number**. Eight tools, each calling the same public API a user could open themselves. Numbers enter an answer only as a tool result, and every answer carries the dataset, the timestamp and the float identifiers it used.



CAP warning areas over the water they concern. 15 bulletins: 12 genuine ones from Indian state disaster management authorities via NDMA SACHET, and 3 ocean-hazard bulletins for tsunami, high wave and swell surge which are **rehearsals we authored**, stamped EXERCISE in the data and on screen. India published no live public ocean CAP feed when this was ingested, and the interface says so rather than letting a drill pass for an alert.



Samudra Sahayak answering "How good is the model?" Assembled from tool results, printing the tool count, the citation list, and the tag planner: **rules**, which says honestly that the planner is a deterministic router rather than a language model. Part 5 explains that choice.

The rest of the product

Six guided tours drive the real scene rather than playing a video, serving the outreach and education mandate the problem statement asks for and doubling as a demo that cannot go wrong. **It runs with the network unplugged:** offline is the default posture, not a mode, so a demonstration can be given with the WiFi visibly off, which turns "deployable on INCOIS infrastructure" into something the audience watches happen. **SagarNode** is a tabletop ESP32 rig with a temperature probe, a turbidity sensor and a TDS probe used as a *conductivity-derived salinity proxy*, which is what it honestly is. It appears on the globe as a live virtual mooring; warm water poured in trips a threshold and fires a CAP-shaped alert flagged as coming from a demo sensor. It demonstrates requirement F6 with hardware instead of a paragraph.

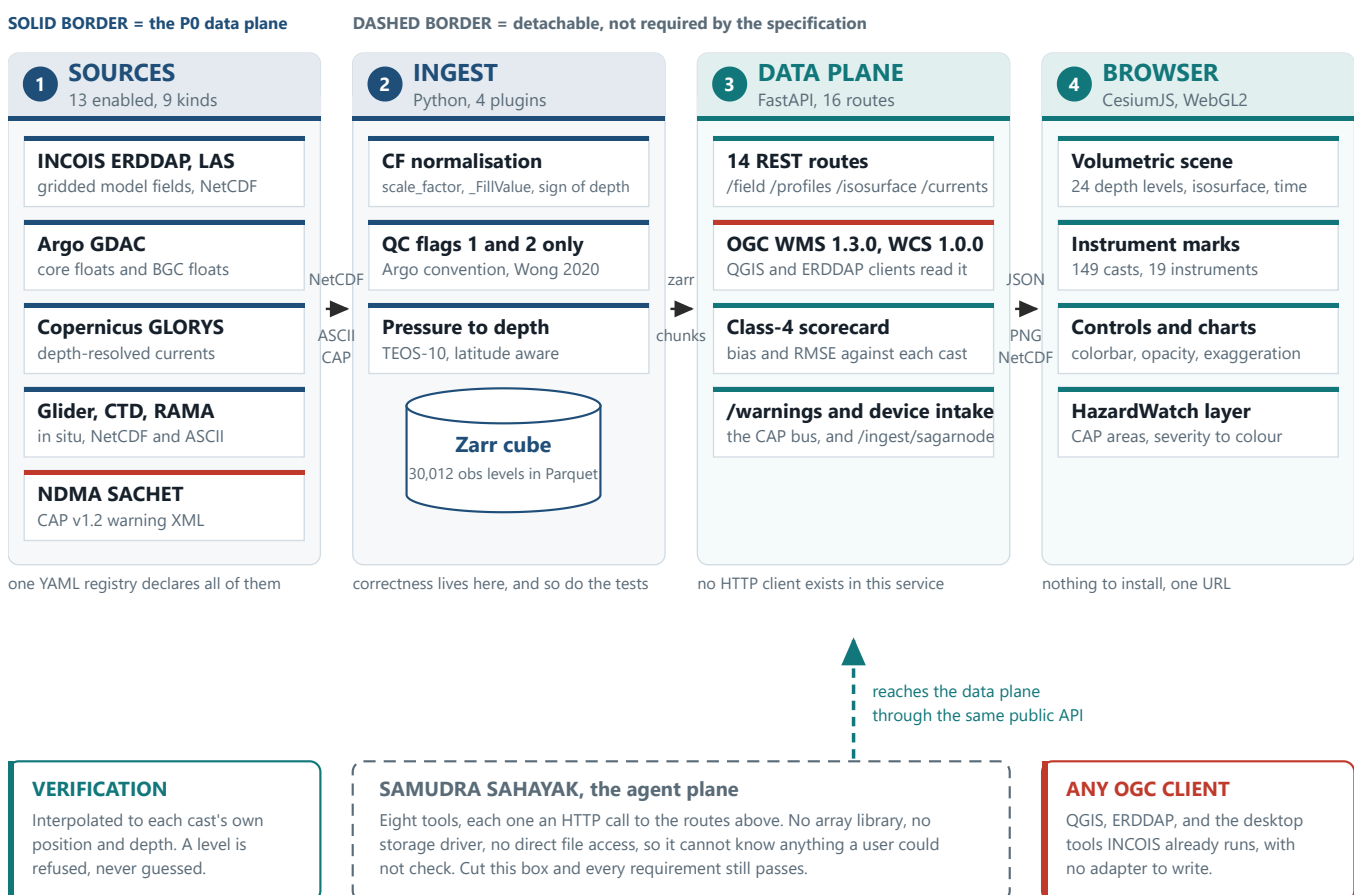
How it works

Everything named here exists in the repository.

4.1 The most important design decision: two planes

The **data plane** is deterministic: files become arrays, arrays become a chunked store, the store is served over HTTP, the browser draws it. It satisfies all seven requirements on its own. The **agent plane** is orchestration, and it reaches the data plane over the same public HTTP API the browser uses, through nothing else.

The split is enforced, not merely described. The agent service's dependency list contains no array library, no storage library and no database driver, so it physically cannot read a data file even by mistake, and therefore cannot know anything a user could not verify by opening the same URL. Cut the agent out and the product still meets the specification.



4.2 Sources and the registry

Fourteen sources declared, thirteen enabled, across nine kinds, all in one file, `data/sources.yaml`. That file is the extensibility surface F3 and F6 ask for: adding a variable or a source is an edit to it, not a code change.

Kind	No.	Reads
erddap_griddap	2	Gridded model fields from an ERDDAP server, the software INCOIS already runs
gdac_geo, gdac_bgc	2	Argo core and biogeochemical float profiles
insitu_tac	2	Copernicus in situ files: the glider and shipboard CTD data
copernicus	1	GLORYS reanalysis, source of depth-resolved currents
mooring	1	RAMA moored buoy, Bay of Bengal
file	2	Plain delimited text: CTD ASCII and ODV spreadsheet exports
cap	2	Common Alerting Protocol warning XML
mqtt, zarr	2	The SagarNode device feed, and the local offline cube

4.3 Ingestion, where correctness is actually decided

Reading scientific files correctly is most of the difficulty, and it is where the tests are concentrated. Four problems recur:

- **CF decoding.** A variable may be stored as small integers with a `scale_factor` and `add_offset` to apply and a `_FillValue` marking absent data. Wrong order, or treating the fill value as real, gives a plausible field that is wrong.
- **The sign of depth.** Some files count depth positive downwards, some negative. Backwards silently inverts the ocean.
- **Pressure is not depth.** Instruments record decibars. The convenient one-decibar-equals-one-metre approximation is wrong by enough to matter: at 1000 dbar at five degrees north, true depth is 992.08 m. We convert through **TEOS-10**, which is latitude aware, and the value is pinned in a test.
- **Quality control.** We keep Argo QC flags 1 and 2 and discard the rest. A subtlety found the hard way: a level whose *pressure* flag is bad is unusable even if its temperature flag is good, because we no longer know where the measurement was taken. That rule is enforced across every parameter.

Output is a **Zarr** store, NetCDF array data re-chunked so a browser can fetch one block over HTTP instead of a whole file, plus a **Parquet** table of 30,012 quality-controlled levels.

4.4 The API, and the four plugins

FastAPI in Python, sixteen routes: `/catalog`, `/field`, `/isosurface`, `/currents`, `/profiles` and `/profiles/{id}`, `/scorecard`, `/warnings`, `/storyboards`, `/plugins` and `/plugins/reload`, `/ingest/sagarnode` and `/sagarnode/healthz`, plus **/wms** and **/wcs**. That last pair is requirement F7 and the strongest deployment argument in the project: because the platform speaks WMS 1.3.0 and WCS 1.0.0, the desktop software INCOIS already runs can read from it on installation day, with no adapter to write.

Plugin	Kind	What it proves
insitu_tac	Source reader	A new observation format can be taught to the platform. Carries five documented defect handlers.
rama_mooring	Source reader	A moored buoy, one of the sensor types F6 names explicitly.
d26_isotherm	Derived field	Depth of the 26 degree isotherm. Nothing measures it; it is computed, then served like any stored variable, including through WMS and WCS.
density_sigma0	Derived field	Potential density anomaly via TEOS-10, with real thermodynamics behind it.

4.5 The browser

Next.js 15, React 19 and TypeScript; **CesiumJS** for the globe, **deck.gl** for custom WebGL layers, **ECharts** for profile plots, and Zustand holding one serializable scene state object that the interface, deep links and the agent all mutate through the same door. Volume rendering follows the performance budget in order: stacked depth-slice textures first because they always work, server-precomputed isosurface meshes second, GPU ray marching last and only where the frame rate holds. Measured on the demo laptop: 60 frames per second at 1920 by 1080 on the integrated chip, against a target of 60 and a floor of 30.

4.6 The agent, HazardWatch, SagarNode, offline

Eight agent tools, each a thin call to a route above: `search_catalog`, `get_field_summary`, `find_profiles`, `get_profile`, `compare_model_obs`, `get_warnings`, `list_storyboards`, and `set_scene`, which is the only way the agent can affect what is drawn. A guard runs over every finished answer regardless of what produced it, and a question outside the tool set gets a refusal listing what the tools can actually do.

HazardWatch turns a parsed CAP bulletin into an event, severity, urgency, certainty, polygon and validity window, drawn as a coloured area with a pulsing border for immediate urgency. The 15 bulletins were reduced from roughly 114,000 polygon vertices to 3,390 by simplification at a hundredth of a degree, which is what makes them cheap enough to draw every frame.

SagarNode is about 230 lines of ESP32 sketch sampling a DS18B20 probe, a TDS probe and a turbidity sensor at one hertz with a ten-sample median filter, publishing JSON over MQTT with an HTTP POST fallback because campus WiFi frequently blocks the MQTT port. It is registered in the same YAML file as every other source. A rolling z-score trips a threshold and publishes onto the same CAP bus HazardWatch reads.

Offline is the default: the `OFFLINE` environment variable defaults to 1, so you opt in to touching the network rather than out. The API has no HTTP client in its dependency list and an assertion that fails if one appears. Exactly one file in the repository reaches the network, `tools/fetch_sample.py`, and the running service never imports it.

What is built, what is not, and the evidence

A status table is only useful if it is willing to say no.

#	Requirement	Verdict	The gap, if any
F1	3D volumetric rendering	Partly	All three named techniques built, six gridded fields render. Chlorophyll is deliberately disabled as a gridded field : the INCOIS series ends in 2020 and cannot honestly share a 2026 time scrubber. It is served as contemporary BGC float data instead.
F2	Instrument data overlay	Met	None. All four named classes live, plus a moored buoy.
F3	Multi-format ingestion	Met	None.
F4	Colorbar and controls	Met	None.
F5	Web-based, scalable	Partly	The Docker path is authored and has never been executed , because Docker is not installed on the build machine.
F6	Extensible design	Met	None. Both extension points have running examples; the device path accepts a plain cur1 as a sufficient sensor.
F7	Open standards	Met	None. Stored and derived variables both served through WMS and WCS.
Beyond the requirement list			
	Class-4 scorecard	Built	Bias +0.026 degrees, RMSE 0.602, 11,718 pairs, worst band 50 to 100 m at 2.07 degrees. It also counts and prints the 4,672 levels it could not use, with a reason for each.
	HazardWatch	Built	15 CAP bulletins parsed and drawn. 12 real, 3 rehearsals marked EXERCISE.
	Guided tours, offline, SagarNode	Built	Six tours and 43 steps; offline enforced by the dependency list; firmware, endpoint, globe glyph and alert path all working.
	Samudra Sahayak	Partly	Tools, schemas, guard, citations and tool trace are all real. The planner is a deterministic router, not a language model.
	Voice, Indian languages	Not built	Bhashini and the speech stack are in the design documents. No speech code exists in the repository today.

THE ONE THING MOST LIKELY TO BE MISREAD

Every answer the question box gives is tagged planner: rules in the interface. A demo that lets an audience believe a language model is reasoning when a regular expression is matching has misled them, and a reviewer who works that out unprompted has caught us rather than been told. The reason it is rules today is practical: an air-gapped demo rules out a cloud model, no local model server is installed, and a small local model doing reliable tool calling is a week of work with a real chance of failing on stage. What matters is that the *shape* is real. The tools are the tools a model would be handed, with the schemas it would be handed, and the guard that stops fabrication already runs on every answer. When a model lands it replaces one function, which means the safety layer will have been in production for weeks before the thing it guards against becomes possible.

Evidence

567	60	11,718	0.602
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tests: 485 API, 71 agent, 11
browser

fps at 1920 by 1080, integrated
chip

model to observation pairs
scored

degrees RMSE, bias +0.026

The test distribution is itself an argument: the heaviest suites are on the data plane, because that is where a silent error produces a convincing picture of the wrong ocean. Plugins 71, text profiles 66, OGC 58, CAP 47, field and isosurface and currents 72, storyboards 30, and 141 across SagarNode, scorecard, density, Argo, CF edge cases, colormaps and offline. Reproduce with `./tasks.ps1 test` and `./tasks.ps1 e2e`; `python tools/validate_rmse.py` re-derives the scorecard.

Three things we found by checking rather than assuming. A counter reporting re-sorted profiles said 475; the true answer was 3, the rest being ascending legs already dropped upstream. A count of levels with no pressure said 20,667; the true answer was zero, because array padding in a ragged file had been counted as missing data. And we recorded F2 as unachievable after searching only Copernicus's rolling thirty-day data part; the archive had the glider and CTD data, the requirement was met, and the reasoning error is written into the status document rather than quietly corrected.

Glossary

ADCP

Acoustic Doppler Current Profiler. Measures water speed at many depths using sound.

Argo float

A drifting robot that sinks to about 2000 m and rises every ten days, recording temperature and salinity. Roughly four thousand active worldwide.

BGC float

A biogeochemical Argo float, adding oxygen, chlorophyll, nitrate, pH and backscatter.

Bias and RMSE

Two ways of scoring a forecast. Bias is the average signed error, so it says whether the model runs warm or cold. RMSE is the typical error size regardless of sign.

CAP

Common Alerting Protocol. An OASIS standard XML format for public warnings, carrying event, severity, urgency, certainty and area. India's backbone for it is NDMA SACHET.

CF Conventions

Climate and Forecast. The agreed rules for how a NetCDF file describes its own contents.

Class 4

The GODAE OceanView protocol for verifying forecasts in observation space, interpolating the model to each observation's exact position, depth and time.

CTD cast

Conductivity, Temperature, Depth. An instrument package lowered from a ship. Salinity is computed from conductivity.

Digital twin

A continuously updated computational model of a physical system, detailed enough to be interrogated in its place.

ERDDAP

A scientific data server exposing the same dataset in many formats over HTTP. INCOIS runs one, and we consume it rather than duplicating its data.

Glider

A winged underwater robot that changes buoyancy to fly up and down in a sawtooth, surveying a chosen line.

HF radar

Shore radar measuring surface currents out to about a hundred kilometres.

INCOIS

Indian National Centre for Ocean Information Services, Hyderabad, under the Ministry of Earth Sciences. Author of the problem statement.

In situ

"In position". A measurement taken by an instrument physically in the water, as opposed to a simulation or satellite estimate.

Isosurface

The surface joining every point of equal value in a volume. The 26 degree isosurface bounds the water a cyclone feeds on.

Marching cubes

The standard algorithm for extracting an isosurface from a 3D grid.

Mooring

An instrumented line anchored to the seabed, reporting from fixed depths. RAMA is the Indian Ocean array.

NetCDF

Network Common Data Form. The standard self-describing binary format for multidimensional scientific arrays.

OGC WMS, WCS

Open Geospatial Consortium standards. WMS returns a rendered map picture, WCS returns the underlying numbers. Speaking both means existing GIS software reads our data unmodified.

Parquet

A columnar table format, efficient to filter and read in parts. It holds our observation levels.

Profile, cast

One column of measurements at a single place and time.

QC flag

Quality control flag, recording how far a measured value can be trusted. We keep 1 and 2, good and probably good.

Reanalysis

A model run over the past with all observations available, giving the best reconstruction of what the ocean did. GLORYS is the Copernicus one.

TEOS-10

Thermodynamic Equation of Seawater 2010. Used here to convert pressure to depth and to derive density.

Thermocline

The depth band where temperature falls fastest, separating the warm surface layer from the cold deep. Our model error is largest there.

Volumetric rendering

Drawing the interior of a 3D body of data rather than its surface or a flat slice.

WMO identifier

The World Meteorological Organization number uniquely identifying an instrument. Ours appear in every citation.

Zarr

A format storing an array as many small chunk files, so a browser can fetch one block over HTTP instead of a whole dataset.

Key references

1. Ryan et al. 2015, GODAE OceanView Class 4 verification. *J. Operational Oceanography* 8. The scorecard backbone.
2. Wong et al. 2020, Argo two million profiles. *Frontiers in Marine Science* 7:700. The QC flag policy.
3. Qin et al. 2021, web 3D framework for ocean forecast data. *Env. Modelling and Software* 135:104908. The closest ancestor.
4. Yu, Qin and Xu 2025, WebGPU volume rendering. *Applied Sciences* 15(5):2782. Early ray termination, adopted.
5. Liu, Silver and Bemis 2019, 3D ocean eddies in browsers. *IEEE Access* 7. The two tier rendering rule.
6. Signell and Pothina 2019, coastal model analysis in the cloud. *JMSE* 7(4):110. The Zarr decision.
7. Tucker et al. 2020, Argovis. *JTECH* 37(3). The co-location API semantics we mirror.
8. Blower et al. 2013, ncWMS. *Env. Modelling and Software* 47. The 2D status quo, and the vertical section idea.
9. Tzachor et al. 2023, digital twins of the ocean. *npj Ocean Sustainability* 2:16. Problem framing.
10. pyParaOcean, IISc Bangalore, EnvirVis 2023 and CGF 2025. Our most serious technical prior art.
11. Khanal et al. 2024, FathomGPT. *ACM UIST*. Pre-register the tools, never generate code on stage.
12. Balakrishnan Nair et al. 2013, INCOIS Ocean State Forecast validation. *Current Science* 105(2).

SagarDrishti, Team PixelPaws. Problem statement SIH26067, Ministry of Earth Sciences and INCOIS. Figures measured on the build machine on 19 and 20 September 2026. Screenshots are unstaged captures of the running build with the network disabled.